

Prompt Ladder – Search & Discoverability ML Task Framing

Baseline Prompt (Weak Prompt)

Prompt

Help me understand this dataset.

Representative Output

The dataset contains information about webpages. You can analyze it using Python and pandas to find useful insights.

Notes

What changed in the prompt:

No changes. This is the baseline prompt.

What improved in the output:

Nothing. The response was very generic.

What still failed:

It did not explain the ML task, target, or how the dataset should be analyzed.

What I would try next:

Add a clear goal.

Version 1 – Added a Clear Goal

Prompt

Help me understand this dataset so I can identify the machine learning task.

Representative Output

The dataset could support different machine learning tasks depending on the target variable, such as classification, regression, or ranking.

Notes

What changed in the prompt:

Added a clear goal.

What improved in the output:

The response focused on identifying an ML task instead of giving a general dataset description.

What still failed:

It still didn't know anything about my project or dataset.

What I would try next:

Add project context.

Version 2 – Added Context**Prompt**

Help me understand this dataset so I can identify the machine learning task.

This dataset is from a Search & Discoverability project using the FlyRank starter dataset.

Representative Output

Since this is a Search & Discoverability project, the problem is likely a ranking task. The dataset may include webpage metrics such as impressions, clicks, CTR, and average position.

Notes**What changed in the prompt:**

Added project context.

What improved in the output:

The response became specific to my internship project instead of listing unrelated ML tasks.

What still failed:

The answer was still unstructured and difficult to copy into my notebook.

What I would try next:

Specify an output format.

Version 3 – Added Output Format

Prompt

Help me understand this dataset so I can identify the machine learning task.

This dataset is from a Search & Discoverability project using the FlyRank starter dataset.

Present the answer under these headings:

- ML Task
- Target
- Success Metric
- Unit of Analysis

Representative Output

ML Task: Ranking

Target: CTR (Click Through Rate)

Success Metric: MAP or NDCG

Unit of Analysis: One webpage per row

Notes

What changed in the prompt:

Added an output format.

What improved in the output:

The information became organized and much easier to use in my notebook.

What still failed:

Some explanations were too technical.

What I would try next:

Define the intended audience.

Version 4 – Added Audience**Prompt**

Help me understand this dataset so I can identify the machine learning task.

This dataset is from a Search & Discoverability project using the FlyRank starter dataset.

Present the answer under these headings:

- ML Task
- Target
- Success Metric
- Unit of Analysis

Explain it for a second-year engineering student with basic machine learning knowledge.

Representative Output

Ranking means arranging webpages in the best order so users are more likely to click the most relevant results. CTR can be used as a target because it measures user engagement.

Notes**What changed in the prompt:**

Added the target audience.

What improved in the output:

The explanation became easier to understand and avoided unnecessary technical terms.

What still failed:

Adding the audience did not significantly improve the technical accuracy of the response.

What I would try next:

Add quality criteria.

Version 5 – Added Quality Criteria**Prompt**

Help me understand this dataset so I can identify the machine learning task.

This dataset is from a Search & Discoverability project using the FlyRank starter dataset.

Present the answer under these headings:

- ML Task
- Target
- Success Metric
- Unit of Analysis

Explain it for a second-year engineering student with basic machine learning knowledge.

Use simple language, include one real-world example, and explain why machine learning is better than using fixed rules.

Representative Output

Ranking predicts the best order of webpages based on multiple factors such as CTR, position, and content type. Instead of always sorting pages by impressions, an ML model learns patterns from the data and recommends a better ranking. This helps users find relevant content more easily.

Notes

What changed in the prompt:

Added quality criteria.

What improved in the output:

The answer became more practical, included an example, and clearly explained why machine learning is more suitable than a fixed rule.

What still failed:

The example was still generic and could be improved with real project data.

What I would try next:

Ask the AI to verify that the response matches the notebook requirements before finalizing it.

Final Reusable Prompt

I am working on a Search & Discoverability machine learning project using the FlyRank starter dataset.

Help me frame the machine learning problem for my notebook.

Organize the answer using these headings:

1. ML Task Type
2. Target or Proxy

3. Success Metric

4. Unit of Analysis

5. Why Machine Learning is Better Than a Fixed Rule

Assume the reader is a second-year engineering student with basic machine learning knowledge.

Use simple language, include one real-world example, explain why the selected ML task is appropriate, and make the ans